Reversibly Stretchable and Tunable Terahertz Metamaterials with Wrinkled Layouts

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Herein, we present a generic strategy for stretchable metamaterials, i.e., wrinkled layout, to reversibly control the structural arrangement of meta-atoms and the corresponding electromagnetic properties of artificial structures. Over the last decade, metamaterials have revealed a hidden capability of artificial structures, from the perspective of manipulating electromagnetic properties, which allow exotic and otherwise impossible functionalities, such as negative refraction, superlensing, cloaking, and perfect absorption.[1–17] In addition to achieving these extraordinary 'frozen' properties, reconfigurable metamaterials (also called tunable or switchable metamaterials) now show promise for providing a new degree of freedom in designing metamaterials with added dynamic controllability. In this regard, various strategies for the reconfiguration of metamaterials have been pursued by utilizing material phase changes, electro-tuning with carbon materials (i.e., carbon nanotubes and graphene), micro-actuating, and mechanical stretching.[18–29] Prominent among these could be a stretchable and flexible metamaterial, especially in terms of its practical applications in a broad range of devices and systems. For example, metamaterials (including nanophotonic systems as well as plasmonics) on a highly stretchable, flexible, and biocompatible substrate, e.g., polydimethylsiloxane (PDMS), could allow for innovations in reconfigurable optoelectronic devices and biological sensors, as has been demonstrated with various stretchable electronic and photonic devices pioneered by the Rogers group.[30–33]

The challenge in producing such stretchable metamaterials is that most of existing conventional lithographic techniques for a rigid semiconductor wafer or a glass substrate are not suitable for plastic or PDMS substrates. For example, the solvents generally used in conventional lithography (i.e., acetone and ethanol) easily dissolve or swell the plastic and PDMS. Moreover, the metals constituting meta-atoms can be cracked easily even under a few strain ratios, which inevitably leads to the degradation of their properties as the number of stretching/relaxing cycles increases.[34] In order to overcome such a process limitation, metal evaporation through a stencil mask[35–37] and the transfer of readily-patterned metals have been considered.[38–41] In addition to these simple approaches, mechanically compliant metamaterials with small feature sizes (such as nanoscale bowties and gratings) have been shown to sustain their structural fidelity during the repeated stretching/relaxing with a certain level of strain.[38,39] Although such works have shown some promise, the possible strain ratio (or tunability) was limited to 23% (the grating and the bowtie),[36,42] and a structural change during stretching/relaxing cycle was clearly observed for a relatively high strain ratio (~50%) (split ring resonator, SRRs).[38,39] Furthermore, stretchable macro-scale metallic structures (tens of micron-scale) for terahertz (THz) metamaterials still face a major challenge in circumventing cracking problems common to all relatively large-scaled metals.[34] Consequently, none has been demonstrated to be suitable for the general platform of stretchable metamaterials.

In this paper, we demonstrate a different platform for reversibly-stretchable and -tunable THz metamaterials, in which all of the macro-scale meta-atoms are wrinkled continuously and conformally over a large area. We show that such a wrinkled configuration allows not only the unprecedented strain ratio of 52.1% without any structural degradation in the 45-micron-scale meta-atoms, but also fully-reversible stretchability and compressibility at least over 100 times. More importantly, the rearrangement of meta-atoms in wrinkled layouts leads to a significant change in the relative transmission by up to 90%. This large tuning capability is attributed to wrinkling-induced resonance splitting and the inhomogeneous broadening of split resonances. Wrinkling-induced tuning of metamaterials can be achieved in a reversible and deterministic fashion with precise variations in the mechanical strain ratio. Moreover, polarization dependent properties can be generated by wrinkling-induced degeneracy lifting in 'polarization-insensitive' metamaterials. With all these attractive features, the generic platform with wrinkled layouts will lead to a new class of stretchable metamaterials that, once again, expands their practical applications.

Our method is based on the 'wrinkling instabilities' that occur when a pre-strained composite layer, consisting of a rigid film and a soft backing layer, is spontaneously relaxed. Following the pioneering work of Whiteside, Hutchinson, and Suo,[43–46] wrinkled layouts have been implemented successfully in stretchable

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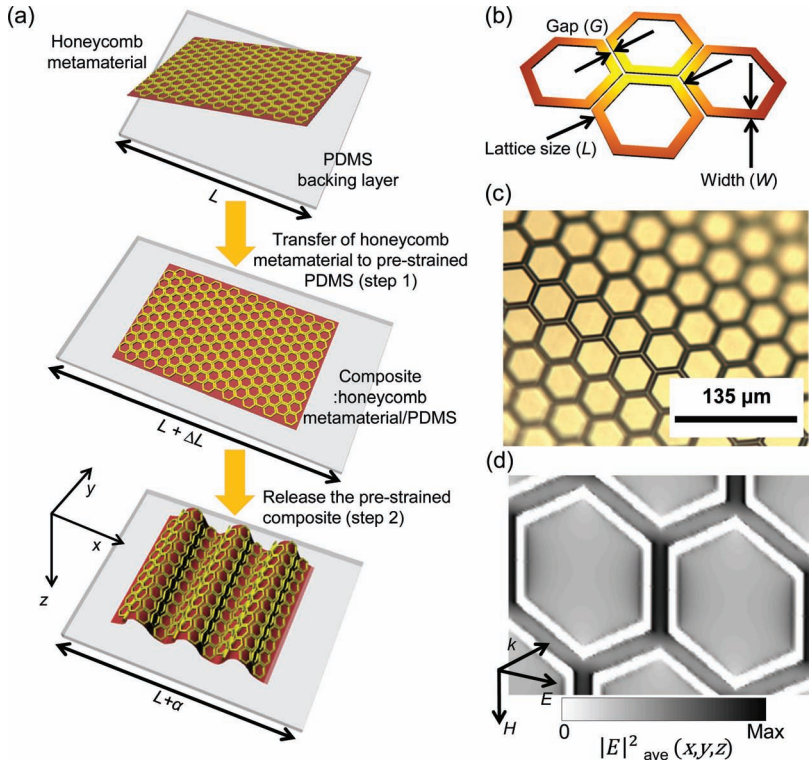


Figure 1. Honeycomb metamaterial and its wrinkling: a) Schematic illustration of the procedure for the fabrication of a honeycomb metamaterial with wrinkled layouts (not to scale). Initial strain ratio was 65% ($\Delta L/L$); but due to the plastic deformation of the PDMS during the heating process, the metamaterial/PDMS composite layer cannot return to its original state (α/L of 12.5%). b) Structural features of the honeycomb metamaterial: gap of 5 μm; lattice size of 45 μm; and width of 3 μm. c) Optical microscope (OM) image of the honeycomb metamaterial (taken in transmission mode). d) Saturated electric field intensity at resonance frequency (1.5 THz).

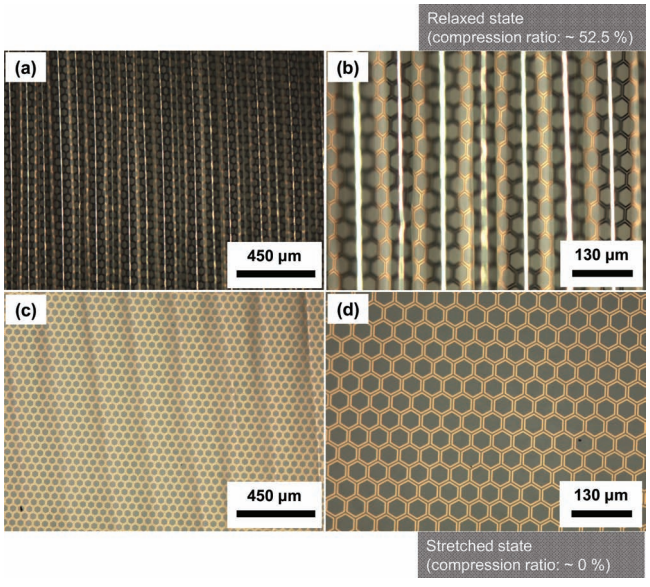


Figure 2. Optical microscope (OM) images of reversibly-wrinkled (a,b) and stretched (c,d) honeycomb metamaterials. OM images of (b) and (d) are magnified images of (a) and (c), respectively. The average wavelength and amplitude of obtained wrinkles are 110 μm and 60 μm, respectively; the strain ratio was 52.5%. All images were obtained in the reflection mode of the OM.

electronics[30–33] and energy devices.[47,48] Furthermore, this kind of wrinkling instability was found to be useful for a convenient evaluation of elastic moduli.[49,50] However, the application of wrinkling instabilities to stretchable photonic systems has yet to be achieved, in spite of several appealing features: (i) reversible stretchability even under high strain ratio (preferably over 50%), (ii) accessibility to a wide range of organic/inorganic materials (extensively used in photonics), and (iii) large and precise controllability of wrinkling features such as wavelength (periodicity) and amplitude (i.e., modulation height). As with the method for other wrinkled forms, wrinkled metamaterials can be made by the relaxation of a pre-strained composite layer (rigid metamaterials and soft PDMS backing layer) (see **Figure 1**a). First, an array of meta-atoms was patterned within a rigid polyimide (PI) layer (guest-host system, Figure S1a) by conventional photolithography (detailed process is described in the Supporting Information, Figure S1); the obtained polyimde (PI) layer (2 μm thickness) containing the array of meta-atoms (metamaterial) was chemically attached onto a mechanically-pre-strained, soft PDMS backing layer (up to 65% of the maximum pre-strain ratio in our experimental condition) by using the transfer printing method (see Figure S2, step 1). Subsequent heating at 180 °C facilitated robust bonding between the silica that had been deposited earlier onto the metamaterial (30 nm thickness) and the preliminarily surface-activated PDMS backing layer with oxygen plasma.[51] In the final step, the heated composite layer (metamaterial/PDMS) was cooled to room temperature (ca. 15 °C), and the subsequent relaxation of the pre-strained composite layer led spontaneously to the formation of wrinkles (step 2, also see Figure S3).

We used periodically-arranged, hexagon-shaped metallic (i.e., gold) frames (uniformly separated by a gap (G) of 5 μm), as a planar metamaterial (that we named "honeycomb metamaterial," Figures 1b-c) in the composite; the width (W), lattice size (L), and thickness of meta-atoms were 3 μm, 45 μm, and 90 nm, respectively. As we previously reported, in this specific design, the effective permittivity at electric resonance can be increased considerably by strong capacitive coupling between adjacent meta-atoms, while the diamagnetic response is effectively suppressed.[52] The electric field is highly concentrated between the meta-atoms, as shown in the calculated electric field intensity profile at the resonance frequency of 1.5 THz (Figure 1d). Thus, the honeycomb design of the metamaterial allows us to probe the influence of wrinkled geometry on the electrical resonance properties of the metamaterials.

Upon relaxation of the pre-strained metamaterial/PDMS composite, the wrinkles were successfully generated, as shown in **Figures 2**a-b. Here, two interesting features are noteworthy. First, the "effective" pre-strain ratio of the obtained honeycomb metamaterial with wrinkled layout (52.5%) was lower than the initial pre-strain ratio (~65%) that was induced initially by

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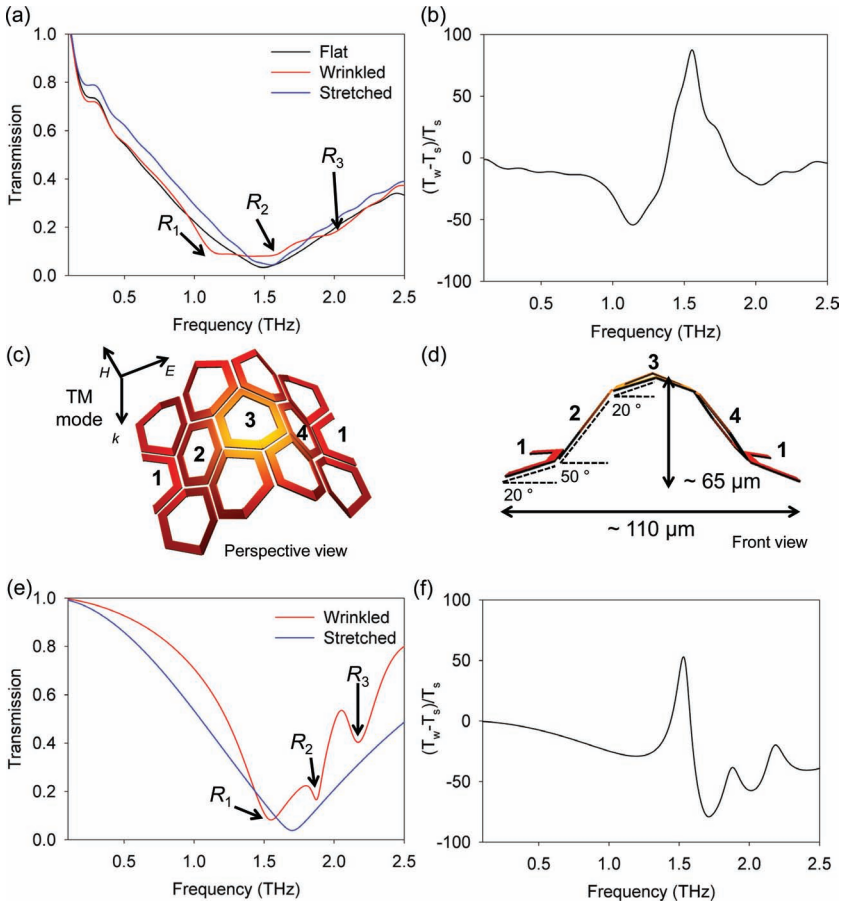


Figure 3. THz resonance properties of a honeycomb metamaterial placed onto a flat PDMS layer, a wrinkled honeycomb metamaterial, and a stretched honeycomb metamaterial: (a) transmission spectra; (b) relative change in the transmission of wrinkled metamaterial, normalized to that of the stretched honeycomb metamaterial; and (c-d) schematic illustrations of the simulated model (four meta-atoms in the direction of a grating vector, each with specific slanted angle (20 and 50 degrees), can be fitted approximately into a single period of wrinkled geometry; (e) simulated transmission spectra; (f) simulated relative change in the transmission of wrinkled metamaterial, which is normalized to that of the stretched honeycomb metamaterial.

mechanical elongation. This is due to the plastic deformation of the polymeric chains of PDMS during the high-temperature heating process (180 °C), which is required for robust bonding between the honeycomb metamaterial and PDMS backing layer;[48] the resulting average wavelength and amplitude (defined here as a peak-to-trough distance) of the wrinkled honeycomb metamaterial were 110 μm and 65 μm, respectively. Second, even when they were in the wrinkled states, the macro-scale metallic features retained its structural integrity, as shown in Figures 2a-b. In particular, the highly-magnified optical microscopic (OM) image of the wrinkled honeycomb metamaterial clearly shows that the meta-atoms are conformally bowed along the sinusoidally wrinkled surfaces (Figure 2b); the meta-atoms at the peaks of the wrinkles were bowed individually without any sign of metal cracking. More importantly, such structural robustness can be maintained during repeated (at least 100 times) stretching/relaxing of the wrinkled honeycomb metamaterials (see Figures S4). Thus, the generic, wrinkled layouts allow metamaterials with exceptional stretchability (~52.5%) and reversibility.

To characterize wrinkling-induced changes in electrical resonance properties and the corresponding transmission spectra of the wrinkled metamaterials, conventional THz time-domain spectroscopy (THz-TDS) was performed; the polarization of the THz wave was set parallel to the grating vector of the wrinkles (TM mode). **Figure 3**a shows the transmission spectra of (i) a honeycomb metamaterial simply attached onto flat PDMS (flat honeycomb metamaterial, black curve), (ii) wrinkled honeycomb metamaterials (red curve), and (iii) fully-stretched honeycomb metamaterials (blue curve). The flat and stretched honeycomb metamaterial clearly showed a resonant dip in the transmission at 1.5 THz, which can be attributed to the electrical resonance of meta-atoms, as has been discussed previously.[52] The experimentally-measured transmission agrees well with the simulations, as shown in Figure S5. In contrast, the overall transmission of fully-stretched honeycomb metamaterial (with a pre-strain ratio of 52.5%) was slightly higher than that of the flat honeycomb metamaterial, while the frequency of resonance remained unchanged. This increase in transmission is due to the reduction in the thickness of the lossy PDMS layer[53] by more than 10% for the fully-stretched composite, while the structural features (i.e., gaps and sizes) and geometries (i.e., hexagonal arrangement) are nearly the same for the flat and fully-stretched honeycomb metamaterials. More importantly, the wrinkling geometry modifies the electrical resonance behavior of the honeycomb metamaterial, as manifested clearly by the changes in transmission spectra near the resonance frequency; note that the single electrical resonance at 1.5 THz evolved progressively into three electric resonances, designated here as R_1, R_2, and R_3 (see Figure 3a). In order to visualize the spectral differences clearly, the relative change in the transmission of wrinkled honeycomb metamaterials, which is normalized to the transmission of fully-stretched state, is shown in Figure 3b; we found that the wrinkling of meta-atoms leads to a significant change of transmission, up to 90%, near the resonance frequency.

The electrical resonances of the honeycomb metamaterials are determined by their structural geometries and the resulting capacitive responses of meta-atoms; consequently, the spatially-varying arrangement of the wrinkled meta-atoms compared with those of stretched meta-atoms naturally leads to both the splitting and broadening (or simply put, inhomogeneous broadening) of the associated electric resonance. Particularly, the wrinkling transforms the two-dimensional, stretched honeycomb metamaterial into a quasi-three-dimensional metamaterial (Figures 2a-b), which is the main cause of the observed

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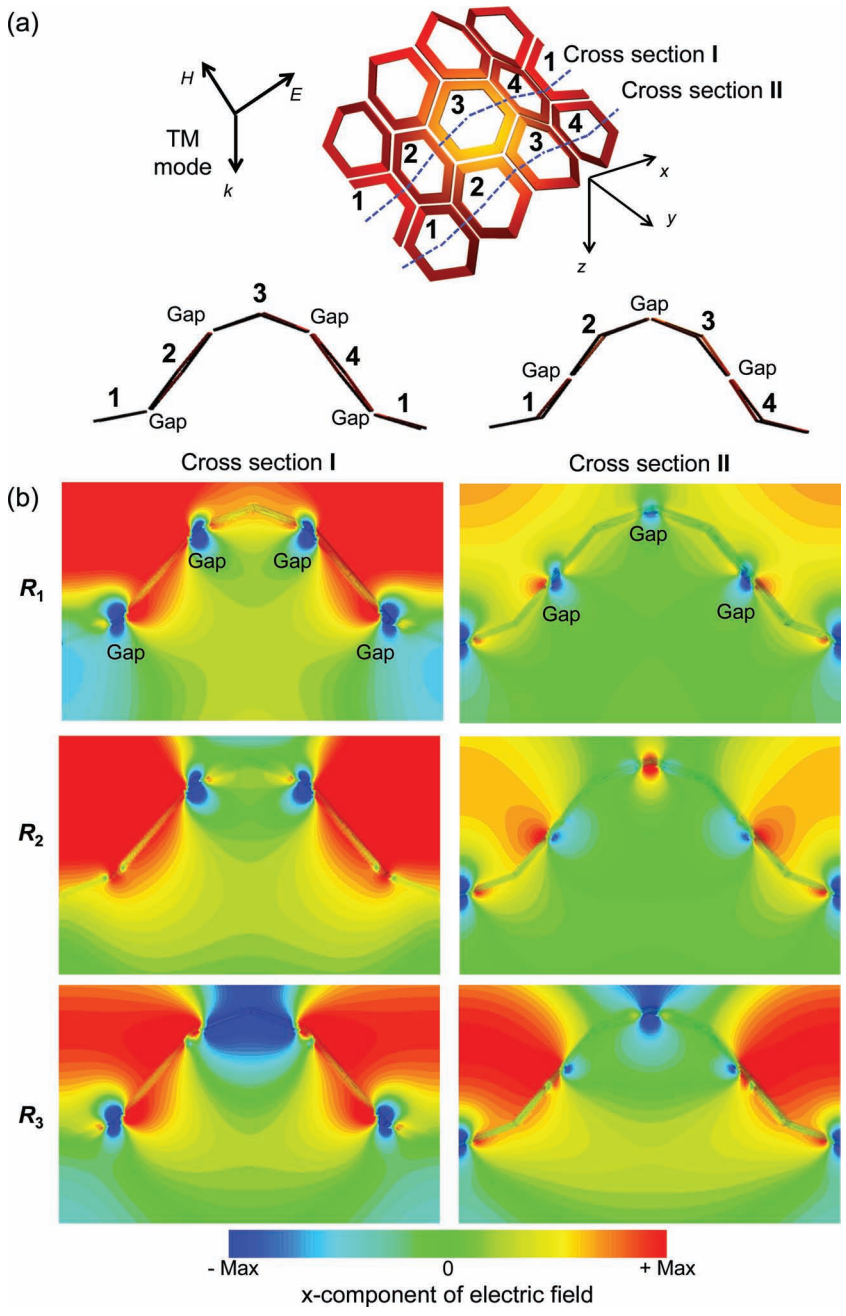


Figure 4. Electric field distribution (E_x) of the wrinkled honeycomb metamaterial at resonance frequencies (R_1, R_2, and R_3): (a) schematic illustration of simulated model (perspective view in the upper part and cross-sectional view in the lower part); (b) simulated electric field (E_x, cross-sectional view).

resonance splitting. In addition, the spatial arrangement of the meta-atoms along the surface of the wrinkles was *partially* randomized. Partial randomness in structural rearrangement of meta-atoms leads to the broadening of split resonances.

To provide further insight into the inhomogeneous electrical resonance broadening that occurred with the wrinkled layout, we approximated the spatial arrangement of the meta-atoms relevant to the geometry of the macroscopic wrinkles (a period of 110 μm and an amplitude of 65 μm) for numerical simulations of the wrinkled honeycomb metamaterial. Figures 3c-d illustrate one such approximated spatial rearrangement of meta-atoms for a single period of wrinkles (also see Figure S6).

Figures 3e-f show the simulated transmission spectra for the stretched and wrinkled honeycomb metamaterial and the relative change in the transmission of the wrinkled honeycomb metamaterial, respectively. Three electric resonances (R_1, R_2, and R_3) are clearly identified in the simulation of the wrinkled honeycomb metamaterial, which is in stark contrast with the case of a stretched honeycomb metamaterial that exhibits single electrical resonance. Note that the resonance frequencies of all simulated results were slightly blue-shifted from those of the experiment due to the absence of the PDMS backing layer in the simulated model. Although a single, specific arrangement of meta-atoms is considered without fully modeling the inhomogeneous reconfiguration of the wrinkled meta-atoms (Figures 3c-d), the numerically-predicted transmission still shows good qualitative agreement with experimental observations. In principle, the experimental transmission would be reproduced in simulation by the superposition of responses from partially-randomized, spatial arrangement of wrinkled meta-atoms. Cross-sectional electric field intensity distributions for the wrinkled honeycomb metamaterials are given in **Figure 4**, which provides a clear pictorial view of the split electrical resonance modes. Three different spatial distributions of electric field (E_x) were clearly observed for three split resonance frequencies, as shown in Figure 4b. Controlled inhomogeneous broadening with wrinkling is distinctively different from the tuning of an electric resonance strength in the SRRs with varying slanted angles, in which the electric resonance frequency was maintained as reported by Averitt et al.[26]

The honeycomb metamaterial belongs to the p6m planar symmetry group, and, therefore, polarization-insensitive responses are expected for this type of metamaterials, which is different from the case of polarization-dependent I-beam metamaterials.[52,54,55] However, the wrinkling of the honeycomb metamaterials led to the polarization-dependent response in the sense that the resonant properties were dependent on the polarization of the THz wave with respect to the grating vector of the wrinkles. **Figure 5**a-b shows the transmission spectra for both the TM and TE modes (see Figure 5c). In the wrinkled state, the strength of the TE resonance is larger than that of the TM resonance, as evidenced by the lower transmission for TE, whereas the resonance behavior of stretched states was approximately the same for both TM and TE modes. In particular, the

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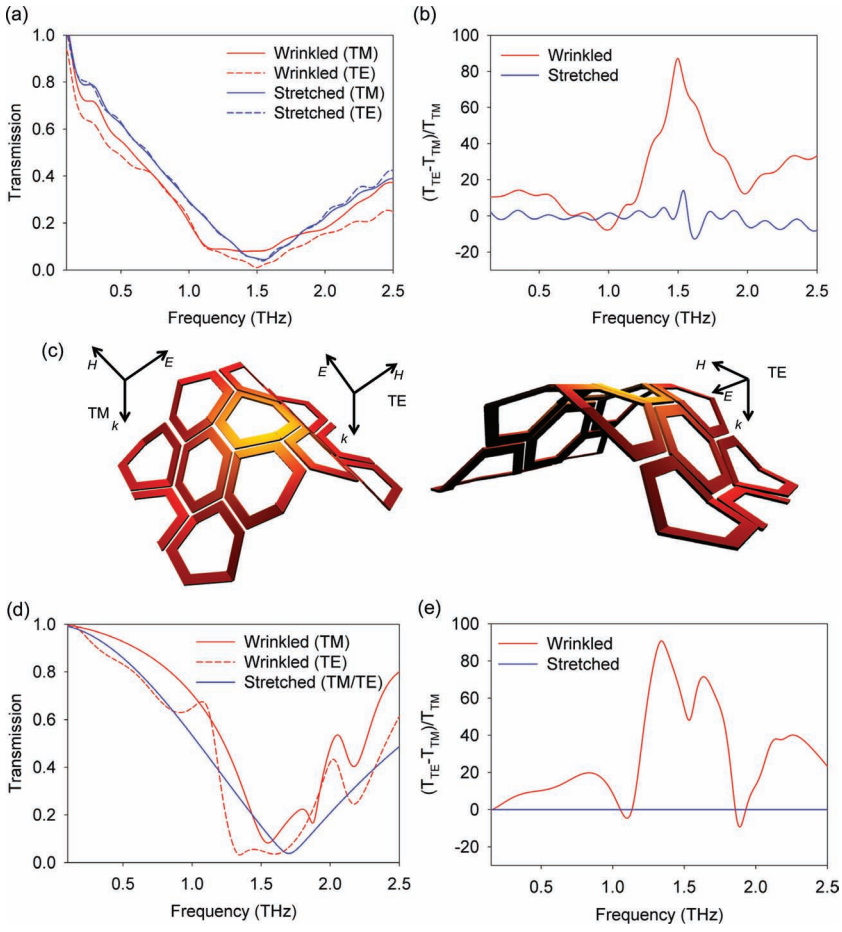


Figure 5. Polarization-dependent properties of the wrinkled honeycomb metamaterial: (a) transmission of wrinkled and stretched honeycomb metamaterials for TE- and TM-polarized terahertz waves; (b) relative change in the TE transmission of wrinkled and stretched honeycomb metamaterials normalized to the TM transmission; (c) schematic illustrations of TE/TM mode excitation for the wrinkled honeycomb metamaterial (left panel) and diamagnetic response driven by H field of TE mode (right panel); (d) simulated transmission of wrinkled and stretched honeycomb metamaterials for TE and TM polarized THz; and (e) simulated relative change in the TE transmission of wrinkled and stretched honeycomb metamaterials normalized to the TM transmission.

transmission ratio of TE to TM was considerably increased near the resonance frequency (up to 90%). This substantial polarization dependency in the wrinkled honeycomb metamaterial has its origin in the diamagnetic response of hexagonal meta-atoms frames, when the TE polarized wave impinges on the slanted meta-atoms (see Figure 4c). The simulated results (Figures 4d-e) qualitatively agree well with experimental observations; some discrepancies are due to the simplified simulated model. As a result, the wrinkling layout also provides a unique route for tuning the polarization-dependent properties of metamaterials, which can possibly be utilized in the implementation of the reconfigurable THz phase retarder or waveplate.

The use of such composite systems, especially those based on the PDMS layer, could bring additional solutions to several key issues, which are closely associated with the reconfigurations and fabrications of mechanically-stretchable metamaterials, especially those with wrinkled layouts. First, as shown in **Figure 6**a, the electrical resonance of our wrinkled PDMS composite metamaterial remained unchanged even after 100 (at least) repeated stretching/relaxing cycles. This remarkable reversibility can be attributed to the intrinsic robustness of the structural integrity of wrinkled meta-atoms (see Figure S4) and to the high mechanical durability of PDMS backing layer. Second, the PDMS-based composite system allows deterministic and reversible control of the mechanical strain ratio of the wrinkled PDMS layer (from 0% to 52.5%), and, accordingly, both the degree and breadth of electrical resonance can be controlled precisely. Figures 6b-c display the experimental results of mechanically-controlled strength and breadth of the electrical resonance of wrinkled metamaterials (for TM mode). An initially-broadened transmission dip of the wrinkled honeycomb metamaterial gradually became narrower by increasing the strain ratio of the wrinkled PDMS layer. This indicates that the structural inhomogeneity was reduced continuously as the mechanical strain ratio of the PDMS increased with stretching. Third, the use of PDMS as a backing layer of the wrinkled layer may also lead to the fabrication of isotropically-stretchable metamaterials (Figure 6d), as well as the one-dimensional, wrinkled structures described above. In particular, the thermoset PDMS could be expanded isotropically upon heating, so that the wrinkling instabilities of the bilayer (consisting of the metamaterial and the thermally-expanded PDMS backing layer) might be spontaneously induced in an isotropic fashion (wavelength of 400 μm and amplitude of 5 μm). Here, as proof-of-concept, the PDMS backing layer was expanded preliminary by 5% (at a heating temperature of 180 °C). Thus, the high robustness and reversibility of PDMS, combined with the deterministic control of the resonance strength and breadth could facilitate the practical applications of these wrinkled metamaterials into THz modulators, waveplates, and biocompatible sensors.

In conclusion, we have demonstrated a simple strategy for fully-stretchable and -reconfigurable THz metamaterial with exceptional tunability. With this generic wrinkled layout, the honeycomb metamaterial can be stretched reversibly up to 52.5% without any degradation in structural integrity. Importantly, the macroscopic wrinkling of the meta-atoms provides a unique way to achieve inhomogeneous resonance broadening. The variation in the mechanical strain ratio allows the simple but deterministic and highly-reversible control of the breadth and strength of THz electrical resonance of the wrinkled metamaterials. Also, the robustness, flexibility, and stretchability in the PDMS backing layer offer excellent motivation for expanding the practical applications of metamaterials, e.g., into

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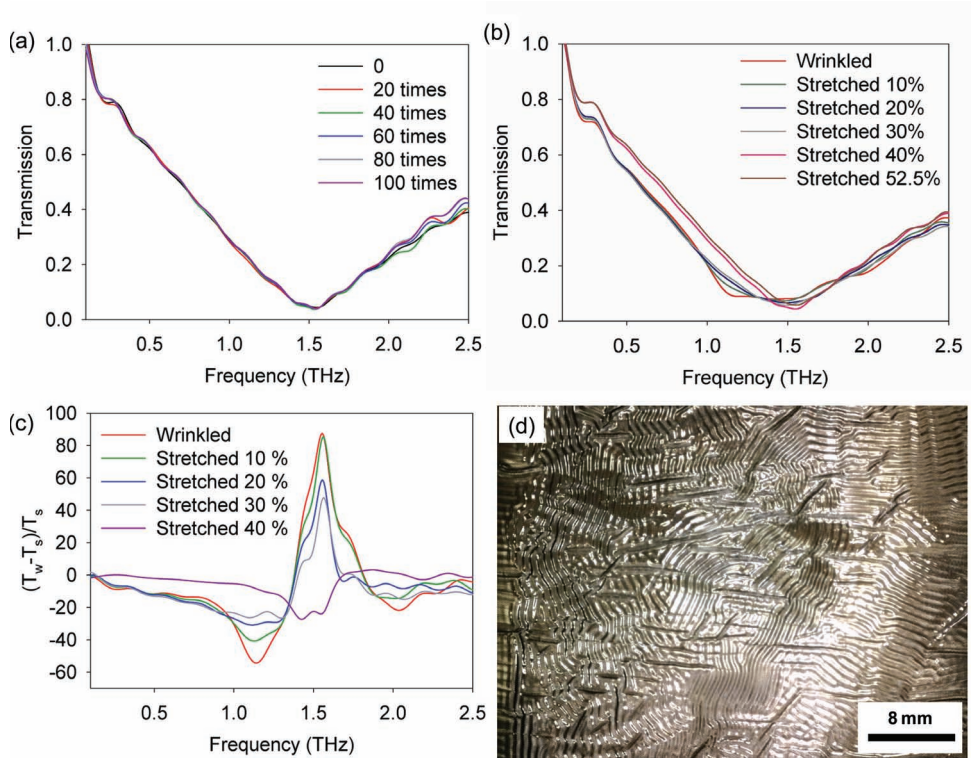


Figure 6. (a) Reversibility of stretchable honeycomb metamaterial with wrinkled layouts: the transmission of stretched metamaterials with repeated stretching/relaxation cycles (100 times). (b) Mechanical control of strain ratio of wrinkled metamaterial and broadening of transmission near resonance frequency (1.1~1.6 THz). (c) Relative changes in the transmission of wrinkled honeycomb metamaterials with varying strain ratios normalized to the transmission of fully-stretched honeycomb metamaterial. (d) Optical image of isotropically (also randomly) wrinkled honeycomb metamaterial.

THz modulators, waveplates, and biocompatible sensors. It is also noteworthy that the wrinkling philosophy is not limited to a honeycomb metamaterial based on electrical resonance; it could be applied readily to a variety of metamaterial systems such as negative index, chiral, and Fano resonance metamaterials. Thus, the wrinkled metamaterial proposed here may offer compelling opportunities for the development of reconfigurable metamaterials that have generic tunable properties, making them promising for use in various practical applications in several fields.

Experimental Section

See the Supporting Information for the fabrication of honeycomb metamaterials, transfer printing, wrinkling process, and a detailed theoretical analysis.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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